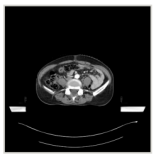
Consistent but Blind? Image Removal Does Not Measure Image Reliance in Medical Vision-Language Models

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Code and data: https://github.com/thaitri2005/Medvlm-Image-Reliance

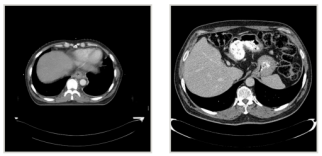
(a) "Is there spinal cord in this picture?" (SLAKE CT, gold: yes)



	with image	2 paraphrases	image removed	quadrant (removed)
MedGemma-4B	yes	yes, yes	yes	Dangerous
Gemma-3-4B	yes	yes, yes	"Please provide the picture."	Ideal

Gemma-3 asks for the image; its forced yes/no choice is "no", so the probe counts it as image-reliant.

(b) "Does the picture contain heart?" (SLAKE CT, gold: yes)



	with image	2 paraphrases	image removed	wrong image	quadrant (removed)	quadrant (wrong)
MedGemma-4B	yes	yes, yes	yes	no	Dangerous	Ideal
Gemma-3-4B	yes	yes, yes	no*	yes	Ideal	Dangerous

** only 51% of first-token probability on yes/no. Same question, opposite verdicts from the two probes.*

Figure 1. **Two real questions, two models, two probes.** (a) Without the image, medically tuned MedGemma-4B still answers, and the consistent-but-dangerous taxonomy calls the answer DANGEROUS; its general-purpose twin Gemma-3-4B asks for the image, and its forced yes/no choice is scored as image-reliant (IDEAL). (b) On the same question, removing the image and showing a wrong image (whose correct answer is *no*) give opposite verdicts for both models. Real model outputs; SLAKE CT images (CC-BY-4.0) [13].

Abstract

The consistent-but-dangerous taxonomy flags a medical vision-language model (VLM) answer as DANGEROUS *when it is stable under paraphrase and unchanged when the image is removed. We test whether image removal measures image reliance well enough to compare models. Three medical VLMs and their general-purpose bases (MedGemma-4B and MedGemma-1.5-4B* vs. *Gemma-3-4B; MedVLThinker-3B* vs. *Qwen2.5-VL-3B) answer 3,365 questions over 907 CT, MRI, X-ray and pathology images with the image removed and under three probes that keep an image in the input: a wrong image, a black image, and an image on which the same question has the opposite answer. Endpoints were pre-registered, and every later probe and model pair was registered before it was run. Image removal finds MedGemma more "consistent but blind" than Gemma-3 (*DANGEROUS *share +13.7 points [10.0, 17.5]). No in-format probe agrees: of nine pair-by-dataset comparisons, image removal separates the models in five, and an in-format probe confirms none of them. Its differences follow refusal instead (forced to answer yes or no without an image, Gemma-3 would rather ask for it on 29% of questions, and its forced answers there are unrelated to its answers with the image) and agreement between a model's priors and the image; MedGemma's excess* DANGEROUS *answers are all correct. The in-format probes need care too: the MedGemmas follow opposite-answer images far*

more than Gemma-3 on SLAKE, which they were trained on (+48 and +37 points), but not on PathVQA, which no medical model was trained on (+0.8 and −5.6; pre-registered). The DANGEROUS *share also rests on a 36–42% chance floor on yes/no questions, and image reliance does not flag wrong answers (AUROC 0.51). Comparing models on image reliance requires in-format probes on data outside their training sets.*

1. Introduction

A model that answers a question about an image without using the image is not doing visual question answering, however accurate it is. In general-domain VQA this concern produced balanced benchmarks and prior-robust splits [1, 8]. Medical VQA is still mostly evaluated by accuracy, where it matters more: a correct answer reached from language priors will be wrong on exactly the patients whose images are unusual.

Accuracy cannot separate *seeing* from *guessing well*, and consistency under rephrasing cannot either: a model that ignores the image is often *more* consistent, because nothing visual perturbs it [20]. Sadanandan and Behzadan [19] therefore crossed the two per sample. *Consistent* means the answer is stable under paraphrase; *image-reliant* means it changes when the image is removed. This gives four quadrants: IDEAL (consistent, reliant), FRAGILE, DANGEROUS (consistent but unchanged without the image) and WORST. On chest X-rays they found that fine-tuning for consistency moves most answers into DANGEROUS.

A taxonomy like this invites comparisons between models, and a comparison is only as good as the probe behind it. Removing the image changes the input format: a model asked about an image that is not there may refuse, answer from priors, or guess, and none of these is the same as ignoring an image that is present. We therefore compare image removal with probes that keep an image in the input, on the same questions and for three medical VLMs, each against the general model it shares an architecture with: MedGemma-4B [21] and MedGemma-1.5-4B against Gemma-3-4B [7], and MedVLThinker-3B against its base Qwen2.5-VL-3B, on CT, MRI, X-ray and pathology questions from three public benchmarks.

Figure 2 summarizes the answer. Image removal reports differences between medical and general models that no in-format probe confirms, in every dataset and pair where it reports one. The in-format probes in turn find the MedGemmas far more image-reliant only on SLAKE, a dataset both were trained on; on PathVQA, which no medical model was trained on, a pre-registered test finds no advantage.

Contributions.

- A side-by-side comparison of four reliance probes on the same questions for three medical/general model pairs and three datasets, showing that image removal's between-model differences are confirmed by no in-format probe. Each probe has precedent [15, 19, 23]; to our knowledge their verdicts about the same models have not been compared.
- Direct evidence for what image removal picks up instead: refusal under a forced yes/no format, and agreement between a model's prior and the image.
- A pre-registered out-of-distribution test showing that two medical models' large in-format advantage on their training distribution does not transfer, so in-format probes also need held-out data.
- Two corrections to the DANGEROUS share: its chance floor on yes/no questions (with Cohen's κ), and a paired split into correct and wrong DANGEROUS answers.
- A pre-registered, open pipeline across three datasets, four modalities and five models, with a paraphrase set screened pair by pair for equivalence and a finding that bf16 batching perturbs log-probabilities by up to 1 nat.

2. Related Work

Language priors in VQA. Balanced pairs [8] and changing-prior splits [1] showed that general VQA models exploit answer priors; text-only baselines are the medical analogue.

Does a medical VLM use the image? PSF-Med [20] measured paraphrase flip rates across medical VLMs on chest X-rays and noted that low flip rates often come with high text-only agreement. The quadrant taxonomy [19] defines reliance by image removal and corroborates it with a KL-divergence score, a black image and random image swaps; on their models these checks agreed with image removal. Zafar *et al*. [23] build blank, pixel-shuffled, absent and CLIP-retrieved hard-negative images to define reliance metrics and train a model against them. Lotfinia *et al*. [15] swap and occlude chest radiographs and find that model class (medical or general) does not predict image use. Pathology VLMs reach high accuracy without the image [25], and medical MLLMs often attend to irrelevant regions [14]. Grounding inputs can even hurt closed-ended accuracy [3]; HEAL-MedVQA [16] and MedHEval [2] benchmark localization and hallucination. These works use one probe or treat several as interchangeable; we compare their verdicts on the same models, and inside and outside the models' training data.

Answering without the evidence. MedVIGIL [11] tests whether models refuse when visual evidence is broken, and Vohra [22] finds that frontier VLMs confabulate diagnoses

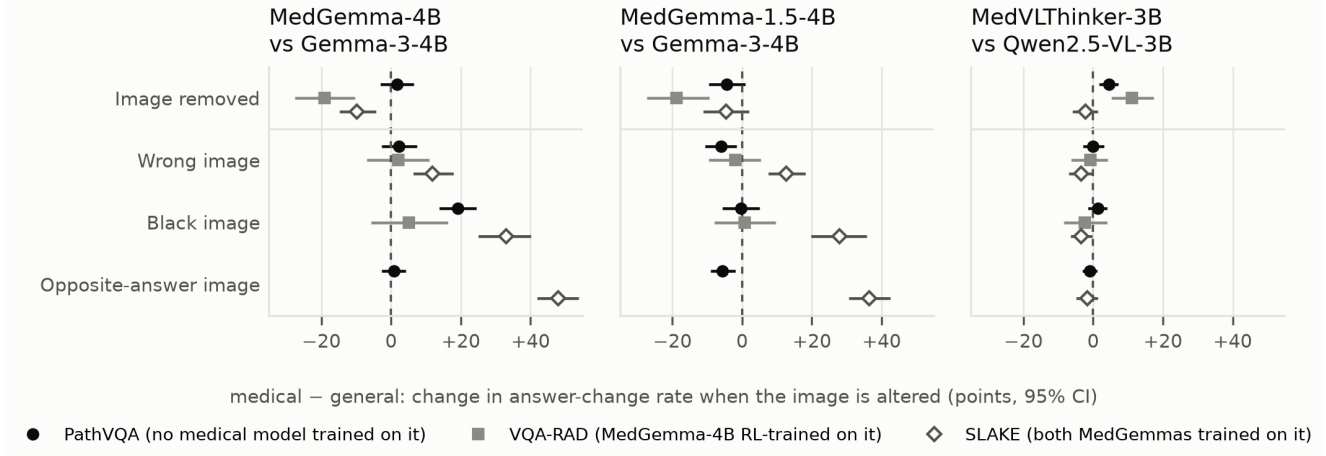


Figure 2. **Which model relies on the image more depends on the probe and on the training data.** Medical − general difference in the share of yes/no answers that change when the image is altered, per probe, dataset (marker) and pair (panel); 95% CIs clustered by image. Image removal (top row) separates the models in five dataset cells; no in-format probe (lower rows) agrees in any of them. The in-format probes separate the MedGemmas from Gemma-3 mainly on SLAKE (open diamonds), their training distribution; the opposite-answer probe on PathVQA (filled circles, registered probe R4) finds no MedGemma advantage.

for images that were never attached. Four of our five small open models decline almost always when allowed to; we show what forcing an answer does to a reliance probe.

Selective prediction. Ranking answers by a score and abstaining on the lowest is evaluated by AUROC and risk–coverage [6]; we use it to ask whether reliance flags errors.

3. Method

3.1. Data

SLAKE [13] (English; CT, MRI, X-ray; 2,114 questions from test and validation, the only dataset with modality labels; validation is added so each modality has 53–82 images). **VQA-RAD** [12] (radiology, clinician-written; all 451 test questions, from the original release). **PathVQA** [9] (pathology, generated from textbook captions; 600 yes/no and 200 open test questions sampled with a fixed seed). A question is yes/no if its gold answer is *yes* or *no*. The **primary analysis set** is the 1,206 yes/no *test* questions on 670 images. Probe R4 uses a separate set: all 1,192 PathVQA test and validation yes/no questions that are also asked about another image with the opposite answer (682 images). Sampling, partner images and image hashes are fixed in committed manifests (SHA-1 seeded), so every run sees identical inputs.

3.2. Models

MedGemma-4B (`google/medgemma-4b-it`) is an open medical VLM whose SigLIP-style encoder [24] and language model were further trained on medical data; its technical report [21] lists SLAKE (450 items, the size of SLAKE's train split) and VQA-RAD (reinforcement-learning stage) among its training data and states that PathVQA was removed. **Gemma-3-4B** (`google/gemma-3-4b-it`) shares its architecture and size. **MedGemma-1.5-4B** (`google/medgemma-1.5-4b-it`, the 2026 release; its card lists SLAKE) is compared with the same Gemma-3-4B, and **MedVLThinker-3B** (`UCSC-VLAA/MedVLThinker-3B-RL_PMC`, Qwen2.5-VL-3B-Instruct trained with reinforcement learning on PMC-VQA) with its base **Qwen2.5-VL-3B-Instruct**. A medical model and its base differ in their whole post-training, not only in medical data, so we compare *checkpoints*. All five run unquantized in bf16 on one 16 GB GPU (quantization would perturb the probabilities we audit) and receive identical prompts: a fixed system line, the question, and "Answer with only 'yes' or 'no'." or "Answer with a single word or short phrase." Each model added after the primary pair passed a registered format check first (mean yes/no first-token mass with the image ≥ 0.9 in every dataset of a 330-question pilot; 0.93–0.999).

3.3. Probes

Every question is posed with its image (**orig**) and with screened paraphrases (**para**$_k$, original image; Sec. 3.5). Reliance is probed by changing only the image or the instruction:

- **Image removed** (*noimg*): image tokens omitted, text identical; the reference probe of the taxonomy [19].
- **Wrong image** (*shuf*): another image from the same dataset and modality, chosen by a hash of the question ID.

- **R1, black image**: a black image in the image slot, in format but without content.
- **R2, "unknown" allowed**: image removed, and the instruction adds "If this cannot be determined without seeing the image, answer 'unknown'."
- **R3, opposite-answer image**: another image of the same dataset and modality on which the *same* question has the opposite gold answer; a model that uses the image must change its answer. Partners exist for 367 SLAKE yes/no questions (130 images), whose templated questions recur across images.
- **R4**: R3 on the 1,192 PathVQA questions above, outside every medical model's listed training data.

The first two belong to the pre-registered main run; R1–R4 and the two added model pairs were each registered after the results that motivated them and before they were run.

3.4. Measures

Yes/no answers come from one forward pass: $P(\text{yes}) = \sigma(\text{LSE}_{t\in Y}\,\ell_t - \text{LSE}_{t\in N}\,\ell_t)$ over the single-token variants of *yes* and *no*; the first-token mass $m = P(Y) + P(N)$ is logged as a format check. **Open answers** use greedy decoding and token-F1 ≥ 0.5 after normalization, with an opposite-term guard (left/right, upper/lower, *etc*. never match). **Consistent**: the answer on the first two valid paraphrases equals the *orig* answer. **Reliant** under a probe: the probe answer differs from *orig*; across probes we compare this *answer-change rate*. The **quadrant** follows from (consistent, reliant). **Refusal**: $m < 0.5$, *i.e*. most first-token mass is on neither *yes* nor *no*; free generation confirms these are requests for the image (Fig. 1a). **Pair accuracy** (R3, R4): right on both images, 0 by construction for a model that ignores the image. **Image PMI** of an answer $\hat{y}$ is $\log p(\hat{y} \mid I, q) - \log p(\hat{y} \mid \varnothing, q)$. **Chance correction**: Cohen's κ [4] between *orig* and probe answers; the *chance floor* of the DANGEROUS share is $P(\text{consistent}) \cdot p_e$, where p_e is the agreement expected from the two answer distributions alone.

3.5. Paraphrase construction and screening

Paraphrase consistency is meaningful only if a paraphrase cannot change the correct answer. Candidates come from Qwen2.5-3B-Instruct [18] (greedy; five styles) and from two deterministic templates that copy every content word ("Is there X?"→"Is X present?"). A rule validator rejects changes to laterality, negation and absence words, quantities, antonyms, imaging planes, question form, answer leakage, near-copies and length. Two audit rounds accepted 81% and 83%, below the pre-specified 85%, so we tightened the rules, excluded a "plain language" style, and **judged every unique rule-valid (question, paraphrase) pair**, 4,890 in total, against a written rubric (Appendix B). The judge was an LLM (Claude); see Sec. 6. 1,208 pairs were rejected, each with a recorded reason (*e.g*. *mass→lesion*, *is enlarged→can be enlarged*), and unjudged pairs are rejected by construction. 84% of the primary-set questions retain at least two valid paraphrases; quadrant analyses use those (1,012 questions, 600 images).

3.6. Statistics and pre-registration

Hypotheses, endpoints, tests and exclusions were frozen before the main run [17]; every later change is listed in Appendix C, and post-hoc analyses are labeled as such. Several questions share an image, so all intervals are percentile 95% CIs from a cluster bootstrap over images [5] ($B = 2000$) with two-sided bootstrap p-values. Holm families [10] (Tab. 2): **P1a–c** (blind accuracy minus the majority-answer prior) and **P2** (AUROC of image PMI minus AUROC of confidence); **H6–H8** (probes R1–R3, MedGemma − Gemma-3); **H9–H11** (added pairs, medical − general: follows R3 more, changes less without the image, refuses less); and **H12a–c** (R4, with a decision rule fixed in advance). **P3**, the paired difference in DANGEROUS share, is a family of its own. A finding is called robust only if its sign holds across a grid of F1 thresholds (0.3/0.5/0.8), paraphrase counts (1/2/3) and "strong" flips.

4. Results

All runs completed (18,896 calls per model in the main run, 3,535 for R1–R3 and 2,384 for R4, five models). Our pipeline reproduces MedGemma's published SLAKE score (token-F1 70.2 [67.5, 72.7] *vs*. 72.3 on the model card; Gemma-3: 45.1 *vs*. 40.2), and the yes/no format mass with the image is ≥ 0.997 in every dataset. MedGemma is the more accurate model with the image (Tab. 1) by 18 points on SLAKE and 25 on VQA-RAD yes/no, both in its training data, and by 2 on PathVQA, which is not.

4.1. The reference probe: MedGemma looks more consistent but blind

Under image removal MedGemma's DANGEROUS share is 50.6/46.8/56.5% (PathVQA/SLAKE/VQA-RAD) against 43.2/28.1/31.0% for Gemma-3 (Fig. 3, left). The paired difference, **P3** $= +13.7$ **points** [10.0, 17.5] ($p < 0.001$; 1,012 questions, 600 images), is positive in every dataset (PathVQA +7.5 [2.6, 12.3], SLAKE +18.6 [12.2, 24.9], VQA-RAD +25.6 [14.7, 36.5]) and in every cell of the sensitivity grid. MedGemma is also the more consistent model (paraphrase flip rate 6.8% *vs*. 13.6% on SLAKE). MedGemma-1.5 shows the same pattern (+9.8 [5.8, 13.9]). Read at face value, the medical checkpoints are more often stable, confident and blind.

4.2. No in-format probe agrees with image removal

Figure 2 puts every probe on one scale, the share of answers that change, and splits it by pair and dataset. Image removal

Table 1. Accuracy (%) on test questions. Yes/no: with the image, without it ("none"), with a wrong image, against the majority-answer prior. Open: token-F1 ≥ 0.5 with the image. CIs are in the repository.

		MedGemma-4B (yes/no)			Gemma-3-4B (yes/no)			Open, with image	
Dataset	Prior	image	none	wrong	image	none	wrong	MedGemma	Gemma-3
PathVQA (pathology)	52.2	65.5	57.2	51.0	63.2	53.5	46.7	10.5	4.0
SLAKE (CT/MRI/X-ray)	50.7	**86.5**	47.6	67.0	68.7	47.6	62.3	66.7	37.1
VQA-RAD (radiology)	53.0	**81.3**	60.6	70.5	56.2	55.4	60.2	46.0	33.5

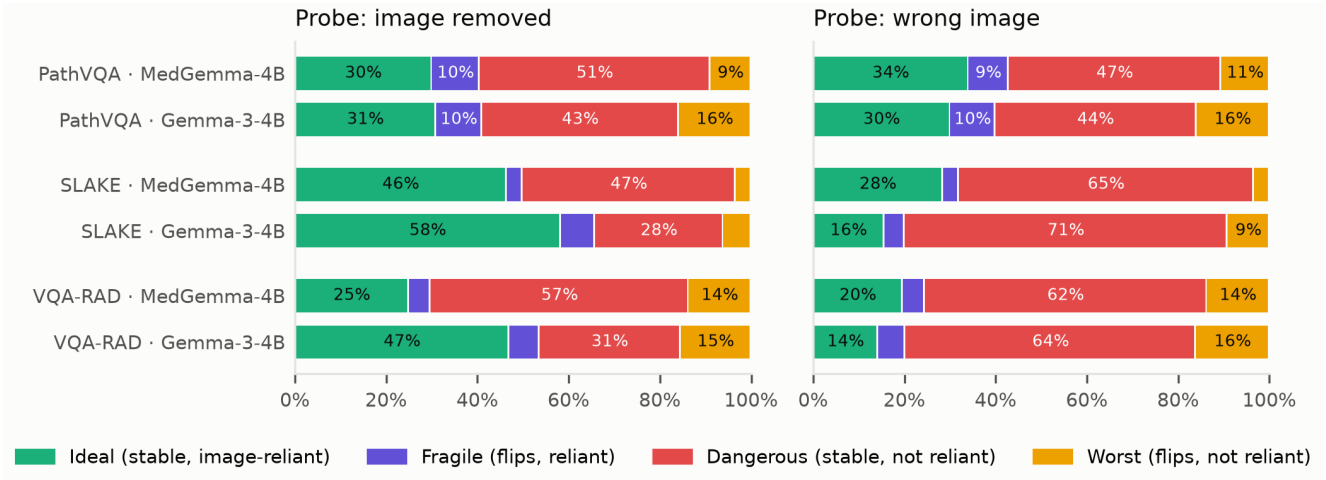


Figure 3. Quadrant composition of yes/no answers (test), MedGemma-4B and Gemma-3-4B. Left: reliance probed by removing the image (the reference definition). Right: by a wrong image of the same modality. Shares are raw; on yes/no questions they include chance agreement (Sec. 4.5) and correct answers (Sec. 4.3).

Table 2. Confirmatory tests. P1–P2 concern MedGemma-4B (P2 on the model's own answer, Appendix C); the rest are paired differences (points, or κ for H6). Pairs: MedGemma-4B *vs.* Gemma-3 (P3, H6–H8, H12a); A: MedGemma-1.5 *vs.* Gemma-3 (H12b); B: MedVLThinker *vs.* Qwen2.5-VL (H12c). 95% CIs clustered by image.

	Test	Estimate	Outcome
Frozen before the main run			
P1a	blind − prior, PathVQA	$+5.0\,[0.9, 8.8]$	p_{Holm}=.042
P1b	SLAKE	$-3.1\,[-7.8, 1.8]$	n.s.
P1c	VQA-RAD	$+7.6\,[0.4, 14.5]$	p_{Holm}=.072
P2	AUROC: PMI − conf.	$-0.20\,[-0.25, -0.15]$	worse
P3	DANGEROUS, image removed	$+13.7\,[10.0, 17.5]$	$p<.001$
Registered later, each before it was run			
H6	κ(image, black) > 0	$-0.22\,[-0.30, -0.14]$	reversed
H7	declines, "unknown" < 0	$-6.7\,[-8.2, -5.3]$	supported
H8	follows R3 > 0	$+48.0\,[41.9, 53.8]$	supported
H9	A / B: follows R3 > 0	$+36.5$ / -1.6	A only
H10	A / B: removed < 0	-7.4 / $+3.9$	A only
H11	A / B: refuses < 0	-28.7 / 0.0	A only
H12a	R4 (PathVQA) > 0	$+0.8\,[-2.7, 4.2]$	failed
H12b	A: R4 > 0	$-5.6\,[-9.1, -2.0]$	reversed
H12c	B: R4 within ± 5	$-0.9\,[-3.0, 1.3]$	supported

separates the medical model from its base in five of the nine cells: lower change rates for both MedGemmas on VQA-RAD ($-19.1\,[-27.6, -10.4]$ and $-18.7\,[-27.3, -9.4]$) and for MedGemma-4B on SLAKE ($-9.9\,[-14.9, -4.4]$), higher ones for MedVLThinker on PathVQA and VQA-RAD ($+4.5\,[1.7, 7.2]$, $+11.2\,[5.3, 17.4]$). In four of these cells no in-format probe finds any difference; in the fifth, MedGemma-4B on SLAKE, every in-format probe finds the opposite (wrong image +11.8, black image +33.0, opposite-answer image +48.0 points). Pooled over datasets the primary pair shows the same split: image removal $-6.1\,[-9.5, -2.7]$, wrong image $+5.1\,[1.7, 8.5]$, black image $+20.3\,[15.9, 24.6]$. The two main-run probes also disagree per question (κ between their reliance labels -0.11 to 0.23), and the modality ordering of the DANGEROUS share reverses between them (X-ray lowest under image removal, highest under a wrong image), partly because a random partner often has the same correct answer (all 23 chest X-ray partners that carry the same question).

The in-format probes on SLAKE. The registered probes R1 and R3 were run on the primary pair after the main results. H6 predicted that MedGemma's answers would sur-

vive a black image more often; it failed in the opposite direction ($\kappa(\text{image}, \text{black}) = 0.11$ *vs.* 0.33): on a black image MedGemma's yes-rate drops from 58% to 28% (it reads an empty image as absent findings), while Gemma-3's rises from 64% to 84%. H8 held: on SLAKE's opposite-answer swaps MedGemma follows the image 72.8% of the time and Gemma-3 24.8%, and MedGemma is right on both images in 71.7% of swaps against 23.4%. MedGemma-1.5 replicates this on SLAKE (H9, +36.5 [30.6, 42.1]); MedVLThinker does not differ from its base under any in-format probe on any dataset except a small black-image difference on SLAKE (-3.4 [$-6.5, -0.3$]). Section 4.4 tests whether the MedGemma advantage holds outside SLAKE.

4.3. What image removal picks up

Refusal. The pre-registered format check ($m \geq 0.9$) failed in exactly one condition, image removed. Free generation explains it: "*Please provide the picture. I need to see the image to determine if there is a spinal cord present*" (Gemma-3). Gemma-3 would rather refuse on 354 of the 1,206 questions, MedGemma-4B on 219 and MedGemma-1.5 on 8 (Fig. 4); MedVLThinker and Qwen2.5-VL never refuse. On refused questions the forced answer is mostly *no* and unrelated to the model's answer with the image ($\kappa = -0.01$ for Gemma-3, 0.04 for MedGemma), so 53–55% of them count as "image-reliant". Image removal's largest differences sit where refusal differs most: on VQA-RAD Gemma-3 refuses on half the questions, 23 points more than MedGemma-4B and 48 more than MedGemma-1.5. Allowed to say "unknown" (R2), MedGemma-4B declines on 93.0% and Gemma-3 on 99.7% (H7, -6.7), so most of the forced-format difference of 11.2 points disappears. For MedGemma-1.5, which refuses on almost nothing, the DANGEROUS gap shrinks to +3.0 [$-3.6, 9.5$] on the 467 questions both models answer.

Agreement between prior and image. Refusal is not the whole story: on the 436 questions where both MedGemma-4B and Gemma-3 answer without the image ($m \geq 0.9$), P3 is still +8.0 [2.9, 13.3], and in the Qwen pair, where nobody refuses, image removal still separates the models. A model whose blind guess matches what the image shows gives the same answer with and without it, and image removal scores that as blind. MedGemma's blind answers beat the majority-answer prior on PathVQA (P1a, +5.0; +11.9 [7.1, 16.6] among questions it answers, on data it was not trained on), and its no-image answers agree with its image answers beyond chance ($\kappa = 0.21$ *vs.* 0.11 for Gemma-3); in the Qwen pair it is the base whose blind answers agree more ($\kappa = 0.16$ *vs.* 0.09), and the base that image removal calls less reliant. The correctness split shows what MedGemma's excess consists of (Fig. 5, bottom): DANGEROUS-and-correct is 39.2% *vs.* 25.1% (+14.1 [11.1, 17.1]), while DANGEROUS-and-wrong, the cell that would mislead a user, is 11.3% *vs.* 11.7% (-0.4 [$-2.7, 2.0$]); MedGemma-1.5's excess is also correct (+9.3 correct, +0.5 wrong).

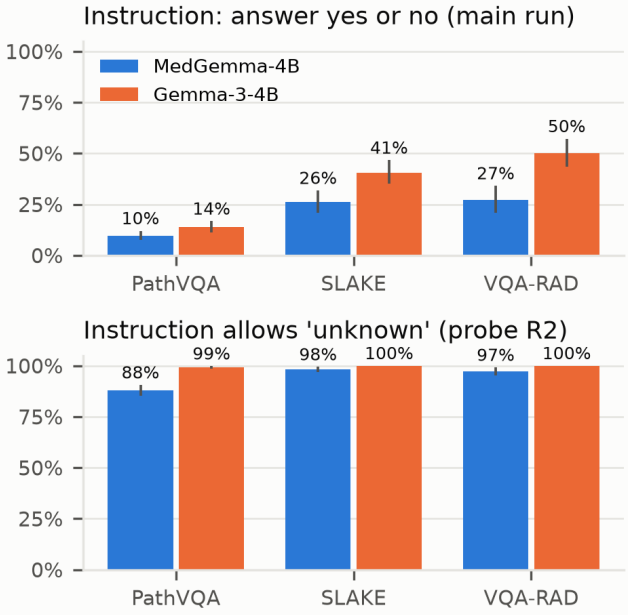


Figure 4. Share of yes/no questions on which the model declines to answer when the image is removed ($m < 0.5$; 95% CIs by image). Top: the main run's instruction ("Answer with only yes or no"). Bottom: probe R2, whose instruction allows "unknown".

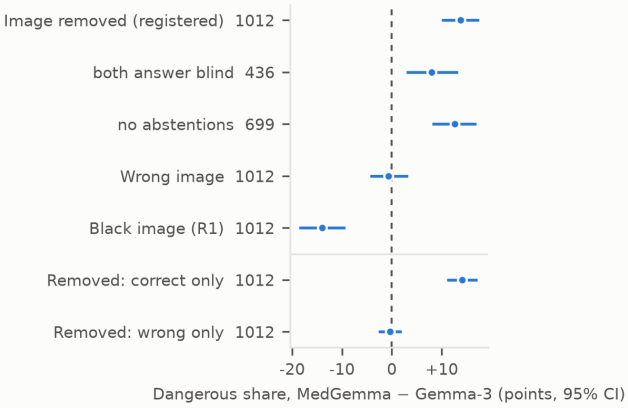


Figure 5. The MedGemma-4B − Gemma-3 difference in DANGEROUS share under each probe and subset, and split by whether the DANGEROUS answer is correct (paired; 95% CIs by image).

4.4. Outside the training data (probe R4)

SLAKE is training data of both MedGemma releases (their evaluated images are disjoint from SLAKE's train split, but the distribution and question templates are not) and not of MedVLThinker, so "trained on SLAKE" explains the

SLAKE results as well as "uses the image more" does. R4 repeats the opposite-answer probe on PathVQA, which MedGemma's training explicitly excluded, for all five models (Tab. 2). Every model follows the partner image on 43–49% of the 1,192 swaps and is right on both images on 36–43%, far from the 0% of a model that ignores the image: all five read pathology images, to a similar degree. MedGemma-4B does not differ from Gemma-3 (H12a: $+0.8\,[-2.7, 4.2]$), MedGemma-1.5 follows *less* (H12b: $-5.6\,[-9.1, -2.0]$, $p_{\text{Holm}} = 0.003$), and MedVL-Thinker matches its base as predicted (H12c). Resampling PathVQA's 154 question templates instead of images gives the same conclusions. Under the decision rule fixed before the run, the SLAKE gap is reported as possibly a training-distribution effect, and the claim that either MedGemma uses the image more than Gemma-3 is withdrawn. This matches accuracy: MedGemma's lead over Gemma-3 is 18 points on SLAKE yes/no and 2 on PathVQA (Tab. 1). On PathVQA the wrong-image and opposite-answer probes find no MedGemma-4B advantage; only the black image does (+19.2), consistent with its reading an empty image as "no findings" rather than with greater use of image content.

4.5. The chance floor

An answer counts as image-independent whenever the no-image answer equals the with-image answer, which on yes/no questions happens often by chance. On the P3 set MedGemma's answers are unchanged without the image 58.6% of the time against 49.1% expected from its answer distributions alone (Gemma-3: 49.7% *vs.* 45.7%). Combined with consistency this gives a **chance floor of 41.5% (MedGemma) and 35.8% (Gemma-3)**, against observed DANGEROUS shares of 50.5% and 36.8%. About 6 of the 13.7 points of P3 are the difference in floors, partly because MedGemma is more consistent (84.5% *vs.* 78.4%), a desirable property the metric counts against it. Corrected for chance, $\Delta\kappa = +0.112\,[0.044, 0.180]$.

4.6. Does image reliance flag wrong answers?

If image-independent answers were the unreliable ones, image reliance could serve as a warning signal. Ranking each yes/no answer by its image PMI separates correct from incorrect answers at chance (Fig. 6; AUROC **0.513** [0.473, 0.552] for MedGemma, 0.532 for Gemma-3), while the model's own confidence reaches 0.709 and 0.603 (P2: better by 0.197 [0.145, 0.250] and 0.071 [0.026, 0.116]). Whether a model leaned on the image does not tell us whether it was right.

5. Discussion

What image removal measures. Removing the image asks a model a question about an image that is not there. Its answer depends first on whether it will answer at all and

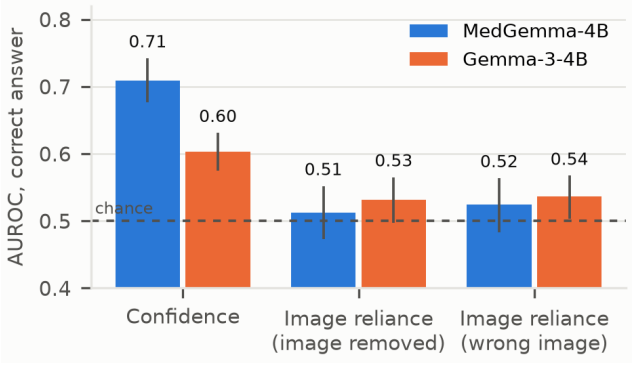


Figure 6. AUROC for ranking correct above incorrect yes/no answers (test, 95% CIs): the model's confidence *vs.* label-free image reliance under each probe.

on how good its priors are. Across three pairs and three datasets, every difference image removal reported between a medical model and its base went unconfirmed by the probes that keep an image in the input, and its largest differences coincided with the largest differences in refusal. The original study's corroborating checks (black image, random swaps) agreed with image removal on its chest X-ray models [19]; on ours they do not. A DANGEROUS share is therefore underspecified without its probe, and a comparison of DANGEROUS shares between models measured by image removal can say more about refusal policy than about image use.

What the in-format probes measure. The in-format probes are not neutral either. On SLAKE they say both MedGemmas use the image far more than Gemma-3; on PathVQA, outside their training data, they say they do not. A model's reliance on the image is a property of the model *and* the data, and in-format probes run on a model's training distribution can reflect what it learned there rather than how it uses images in general. For these three pairs, no general claim that the medical model relies on the image more survives held-out data.

Where the residual risk is. MedGemma's extra DANGEROUS answers are correct, and the stable, image-independent and wrong cell is the same for both models. MedGemma's errors, fewer overall (25.7% *vs.* 37.6%), are more often of this kind (44% *vs.* 31%, +12.9 [6.0, 19.9]), so a consistency or reliance check would miss them.

Recommendations for reliance audits. (1) Probe reliance in format, ideally with an image that calls for a different answer, and report agreement between probes. (2) Run in-format probes on data outside each model's training set,

and check training documentation before comparing models. (3) Report the DANGEROUS share with its chance floor or as κ; on balanced yes/no data a model whose blind answers are pure noise is about 50% "not reliant". (4) Split DANGEROUS by correctness. (5) Allow abstention, and treat declining without an image as desirable rather than as a format failure.

6. Limitations

Three pairs from two model families show that image removal *can* disagree with in-format probes and suggest why; they do not estimate how often across the field. The refusal and prior-agreement accounts and the per-dataset breakdown (Fig. 2) were formed after the primary results; of the dataset-specific comparisons only R4 was registered. R4 covers one modality, pathology, on which all models are weak (pair accuracy 36–43%), so it argues against a general advantage but cannot rule one out in every domain. R3 and the SLAKE results rest on templated questions that recur across images; resampling templates instead of images leaves the conclusions unchanged. MedGemma-4B was also trained on VQA-RAD, whose test images almost all occur in its train split, so its VQA-RAD results may reflect exposure; the general models' training data are not published. Checkpoints differ in their whole post-training, not only in medical data. Refusal is inferred from first-token mass and confirmed on a sample by free generation. Paraphrase equivalence was judged by an LLM and **was not validated by independent human raters**; a second check of 50 used paraphrases, rated by the author with AI assistance, accepted 88% (above the pre-specified 85%), but it is not independent of AI judgment. This is the main limitation of the consistency measure; the probe comparison does not use paraphrases. PathVQA questions are auto-generated and often ungrammatical. The chance correction, correctness split and refusal split were added after seeing the results. Excluding the 330 pilot questions used to tune the design leaves every conclusion unchanged (P1a +6.3 [2.4, 10.6], P3 +12.9 [8.9, 17.1]). Benchmark VQA is not clinical use, and nothing here is a clinical claim.

7. Conclusion

Is a small medical VLM consistent but blind? By the image-removal test MedGemma looks that way, more than its general-purpose twin. No probe that keeps an image in the input agrees, in any dataset or pair we tested: image removal measures whether a model answers without an image and whether its priors match the image. Probes that keep the image find the medical models more image-reliant only on data they were trained on. Comparing models on image reliance needs in-format probes on held-out data, chance correction and a correctness split.

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A. Implementation details

Pipeline. A Python package (`medvlm`) with a command-line interface: `prepare` (download, deterministic sampling, SHA-checked images), `paraphrase` (generation plus rule and judgment validation, fail-closed), `pipeline` (resumable inference in parts with checkpoint analysis), `probes` (R1–R3), `contrast` (R4), `analyze`, `compare`, `posthoc`, `replication` and `contamination` (all tables and figures here). Configs layer base, profile, model and hardware settings. Each job is one (question, condition) pair keyed by ID, so interrupted runs resume without recomputation, and every run records its config, commit and library versions. Registrations are git tags.

Inference. PyTorch 2.11 and Hugging Face Transformers 5.19, bf16, SDPA attention, one RTX 4080 SUPER (16 GB). MedGemma-4B runs at 0.23 s per call (18,896 calls in about 75 minutes, 8.2 GB peak). During pre-flight we found that in bf16 the *batch size* changes teacher-forced log-probabilities by up to 1.09 nats (median 0.17) even with identical, unpadded inputs, while $P(\text{yes})$ changes by at most 0.008. An fp32 output head did not remove it, so the cause is inside the network. All runs therefore use batch size 1, which costs about 1.5× throughput. A regression check reproduced the T4 pilot on the new GPU (98.5% identical answers, mean $|\Delta P(\text{yes})| = 0.004$).

Testing. 103 unit tests and 7 real-model tests cover, among others, scoring edge cases, paraphrase rules (including each failure class found in the audit), batching equivalence, crash-safe resume, the coverage of the clustered bootstrap (by simulation), the test-split restriction of P3, chance agreement of κ, the construction of opposite-answer partners and an oracle-*vs*.-blind check of pair accuracy.

B. Paraphrase equivalence rubric

A pair is **accepted** only if it asks exactly the same thing, so the correct answer cannot differ. Allowed: perception framing ("can X be seen", "is X visible", "can X be seen to be enlarged"), exact synonyms (clavicle/collarbone, encephalon/brain, intravenous contrast/dye), reordering. **Rejected**: a fact turned into a possibility or capability ("Is the heart enlarged?"→"Can the heart be enlarged?"); a broadened or narrowed term (mass→lesion, organs→structures, digestive system→GI tract, skull fracture→cranial injury); a changed scope or presupposition ("Do the organs exist in X"→"Are there organs in X"); added qualifiers, findings or modality ("bilaterally", "on imaging studies", "radiograph"); a reversed or changed relation ("does A show B"→"is A associated with B", "mass lateral to X"→"X lateral to mass"); antonyms (healthy↔unhealthy, coronal→transverse); two questions in one; malformed text. Rejections by class: relation 447, term 364, modal 106, added content 102, organ→structure 47, generic collapse 30, polarity 19, scope 9 (plus template-pass rejections).

C. Deviations from the pre-registration

Every change made after the analysis plan was frozen, with its reason.

Change	Reason and effect
Batch size 1 for all runs	bf16 batching shifts teacher-forced log-probabilities by up to 1.09 nats; decided before the main run.
Validator v2/v3, *plain* style excluded, full equivalence screen, template styles	Audit acceptance below the pre-specified 85% line; before the main run.
Format stop-the-line rule triggered (image removed only)	Model behavior (refusal), not a pipeline error; forced-choice $P(\text{yes})$ kept, refusal reported, P1/P3 re-estimated without refusals.
P2 computed on the model's answer $\hat{y}$	The registered formula used the gold answer, which is unavailable at deployment and predicts correctness by construction.
P3 restricted to the test split	An analysis bug had pooled the validation split; fixed to the registered set.
Chance correction, correctness split, refusal split, answer-change comparison across probes, per-dataset breakdown	Added after the run; descriptive, reported next to the registered estimates.
Robustness probes R1–R3	Registered after the main run, before running them (pre-registration §15); R3 restricted to SLAKE as registered.
Added pairs, H9–H11	Registered before running (§16); all three new models passed the registered format check.
Probe R4, H12a–c	Registered before running (§17) after the training-data check; H12a not supported, H12b reversed; the registered decision rule withdrew the claim that the MedGemmas use the image more.
Training-data statements	Corrected after re-checking the documentation (§18): MedGemma-4B was trained on VQA-RAD; MedGemma-1.5's card lists only SLAKE.